

Rules and Regulations for **Lego Line** Category

1. About Competition

The Lego Line category is a robotics competition where participants design and program autonomous robots with only Lego parts to follow a track and complete it in the shortest time. This category encourages creativity, technical knowledge, teamwork, and problem-solving skills while promoting fair play and innovation. It provides students with an exciting opportunity to explore robotics and engineering in a competitive environment.

2. Robot Specification

1. Materials and Equipment:

- Robots must be built entirely from LEGO components. Custom-made or 3D-printed parts are strictly prohibited.
- Any LEGO wheels or tracks are allowed, including those from EV3, Spike, or LEGO Technic sets.
- Only one active LEGO controller (brick) may be used during the match, such as NXT, EV3, Spike, or 51515. A second controller may be used for building or programming purposes, but not during competition.
- There is no limit to the number of motors or sensors used, as long as they come from LEGO robotics kits like NXT, EV3, Spike. LEGO Technic motors that are not part of these kits are not permitted.

2. Size and Weight:

- The robot must not exceed a 30 cm × 30 cm boundary.
- The robot must not exceed a maximum weight of 3kg.
- No height limit.

3. Power and Signal Communication :

- The robot must work autonomously by power source.
- Wi-Fi and Bluetooth must be turned off during the match. The program must be uploaded before the match.

3. Match Regulations

1. Start:

- The competition consists of 3 rounds. The next round begins after all teams have completed the current round.

2. Valid Result Time:

- Each team has a maximum of 3 minutes to complete the line following run. Teams that do not complete the run within the time limit will be recorded with a time of 03:00:01.
- Any penalties incurred during the run will be added to the final time.

3. Victory Conditions:

- The winner will be determined based on the fastest single run out of 3 total rounds.
- Teams will be ranked according to their best official time (including any penalties).

4. Match Procedure:

- Connecting to the robot via Bluetooth or Wi-Fi is prohibited during matches. If anyone connects to the robot via Bluetooth or Wi-Fi, the team will be disqualified.
- The robots must clearly display an identification number assigned by the organizers on the exterior casing in a clearly visible position.
- Once the robot crosses the start line, it must operate fully autonomously. Otherwise, it will be disqualified.
- If the robot's entire vertical projection leaves the line, goes backward, or stops, it is considered off-line. The participant must place the robot back where it left the track. Each violation adds a 2-second penalty to the final time.
- If 3 off-line violations happen on the same round will round be considered fail, resulting in no valid result.
- The referee has the sole and final authority to decide on off-line violations.
- If a round is considered a fail, it will not be repeated, and no time will be recorded.

5. Repairs and Maintenance

- If a robot experiences a failure during a match, the round will be considered as fail.
- Teams are allowed to repair, test, or modify their code and robots during the breaks between rounds.
- A competitor may request a stoppage of the match in the event of personal injury or significant robot damage that prevents the match from continuing. A break will be granted if the referee determines it is necessary and justified.

4. Sportsmanship and Fair Play

1. Sportsmanship:

- Participants are expected to demonstrate fairness, respect, and good sportsmanship throughout the competition.
- Unsportsmanlike conduct may lead to disqualification. The Referee reserves the right to disqualify any supervisor, participant, or robot found to be acting unethically or in violation of sportsmanship standards. All decisions made by the Referee are final and not subject to appeal.

2. Inspections:

- Before the competition, the robots will be inspected to ensure they comply with rules
- Referees may ask you about the robot to ensure that the student built the robot by itself. Referees may ask you before or after the competition.
- The robot must pass inspection before the competition. If the size or weight exceeds the limits, a maximum of 5 minutes is allowed for structural modifications. If it still fails inspection after this time, the competition qualification will be revoked. The referee will time this period with a stopwatch.

3. Participant behaviour:

- Participants are not allowed to touch the robot during the round, except when off-line violation happens.
- During the game, 1 participant and 1 assistant must be present during the game. Other participants cannot be with them.

4. Maintenance:

- Participants may repair the robot before the match starts.
- Mentor and Assistant can help you and even give you advice during the game.
- If there's a problem with the robot during the match, the referee will give you 2 minutes to fix it. A mentor or assistant can help you.

5. Authority of the Judges:

- The Referee has the authority to supervise all matches and ensure that the competition rules are followed correctly by all participants.
- The Referee has the power to make final decisions regarding victories, disqualifications, rule violations, and match results during the competition.
- The Referee is responsible for maintaining fairness, safety, and discipline throughout the event and may stop a match if necessary.
- The Referee may provide warnings or penalties to participants who do not follow the official regulations and competition procedures.

5. Safety Rules

1. Safety of Robot:
 - The robot must not have any parts that could harm the participant or track.
 - The referee will check your robot for safety, and if it's accepted, you'll be able to participate. If not, you'll have time to fix it.
 - Participants are forbidden from using any device that can lift a robot from the ground.
2. Approval:
 - All robots will be checked for safety to ensure compliance with the rules.
 - If any safety issues are found with the robot, participants will have time to fix it or it will be disqualified.

6. Track Specification

1. The line follower field mat size is 3m wide by 4.5m long.
2. The field surface is white. The track is a black line with a width of 15mm.
3. The track has a start area, and the track ends with a finish area. The start area and finish area are the same overlapping zone.
4. There may be intersections(crossing perpendicular lines), zigzags.
5. There may be 90 degree turns.
6. The distance from the line to the nearest field edge shall be no less than 15cm, measured from the center of the line

7. Disqualification

1. The robot will be disqualified if the robot does not comply with the rules.
2. The robot will be disqualified if it damages the track or participant.
3. If a participant shouts or goes against the referee.
4. If a participant will disrespect other teams.
5. If the participant won't come after the call-up.
6. If you are called to play and you don't come in 5 minutes the robot will lose the match